

Vibhor Vanvani

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WORK EXPERIENCE

D. E. Shaw

Hyderabad, India

Financial Research Analyst, Data Analytics (promoted from Associate in 18 months)

Jun 2024 -- Present

Analytics support to traders, portfolio managers, and senior researchers across quantitative research, LLM tooling, and investment decision workflows.

- **LLM Engineering:** Built custom LLM-powered tools -- chatbots, OCR pipelines, data parsing and classification systems, and API-integrated extraction across 2,000+ broker reports and earnings calls; reduced manual research effort by ~2 FTE; won D. E. Shaw's internal international prompt engineering challenge.
- **Automation & Dashboards:** Engineered Python HTML dashboards and security trackers monitoring 7,000+ APAC equities across futures, short exposure, and derivatives; automated alternative data and earnings processing pipelines.
- **EM Credit Research:** Developed short-term exit timing forecasts for 6,000+ emerging market bonds across 50 countries via Bayesian regression on Bloomberg and FactSet data, improving accuracy by 70%; supported a \$75M activist position returning 103% in six months.
- **Corporate Development:** Assessed market entry feasibility and portfolio adaptability for new APAC investment theses -- liquidity analysis, regulatory impact modelling, and positioning for firm-level strategy shifts.

EDUCATION

University of Cambridge

Cambridge, United Kingdom

MPhil in Economics and Data Science (Incoming, Oct 2026)

Coursework: Machine Learning in Economics, Causal Inference & ML, Econometric Methods, Microeconomics, Research Computing, Empirical Finance

Recipient of the Cambridge Trust International Scholarship for academic excellence

University of Delhi: Ramjas College (NIRF Rank 5)

Delhi, India

BA (Hons) Economics | GPA: 8.095/10.0 | Top 10 of 100

Nov 2021 -- Jun 2024

Coursework: Econometrics, Data Analysis, Statistics, Linear Algebra, Microeconomics, Macroeconomics

Named to the Dean's Honour List for exceptional academic performance.

Indian Institute of Technology Madras (NIRF Rank 1)

Chennai, India

Diploma in Data Science and Applications | GPA: 8.02/10.0

Sep 2021 -- Dec 2023

Coursework: Machine Learning, Business Analytics, Programming in Python, Statistics for Data Science

Awarded Best Capstone Project in Business Analytics (2023)

INTERNSHIPS

Ernst & Young -- Data Analytics Intern

Delhi, India | Jun -- Aug 2023

- Applied ML models (Random Forest, Gradient Boosting, SARIMA, Prophet) to trade-flow data across 50+ countries; improved forecast accuracy by 25%; insights shaped national policy on market access and product diversification.

Institute of Economic Growth -- Research Intern

Delhi, India | Feb -- Mar 2023

- Surveyed 80+ households on anaemia vulnerability; mapped regional hotspots; contributed to policy recommendations on hemoglobinometers in Anganwadis and iron/folic acid in Mid-Day Meals.

Emirates: National Bank of Dubai -- Market Analytics Intern

Dubai, UAE | May -- Jun 2022

- Selected from India for a Dubai role; analysed uptake of banking and Shariah-compliant products via segmentation and pricing; recommended changes improving product positioning and acquisition strategy.

RESEARCH

V. Vanvani, "[Examining the relation between Calorie Deficiency and Income in Rural India.](#)" The Economics Journal, SRCC, 2022 -- NSSO data across 11 income classes; weak income-nutrition correlation (0.21) exposing behavioural gaps in food expenditure.

V. Vanvani, "[Asset Pricing in an Interstellar Economy](#)" (SSRN) -- Asset pricing model for delayed-information economies; cointegration tests show persistent price dispersion and a volatility-driven risk premium.

SKILLS

Technical: Python, R, MATLAB, Mathematica, Excel, Tableau, Power BI, Bloomberg Terminal

Certifications: Google Data Analytics Professional Certificate (Coursera, 8-course: spreadsheets, R, Tableau)

PROJECTS

[Espresso Protocol](#) -- Agentic econometric AI engine: takes a plain-English question and dataset, selects the correct regressor, tests assumptions, auto-corrects on violations, and returns verified results. All computation is deterministic. Terminal CLI with REPL, HTML dashboard, and LaTeX export.

[LLMs in a War of Attrition](#): Built a multi-agent game theory simulation across four U.S. debt-ceiling crises on live real time macro, bonds, and congressional inputs; Quantified bargaining dynamics using regression analysis, showing that relative delay costs strongly predicted concession timing.

Democracy & Carbon Emissions (DU): Multiple regression across nation-states; stronger democratic institutions correlate with lower emissions growth.

POSITIONS OF RESPONSIBILITY

Debating Society: Secretary: Mentored 70+ members to 10+ national wins; founded college's first Debate Equity Committee (+30% participation).

Ramjas Economic Review: Evaluated 30+ economic research papers using data-driven critique, reducing average review cycle from 6 weeks to 2 weeks

Economics Society: Led large-scale economics events with 1,000+ participants, managing end-to-end logistics and budgets

Satcom / ISRO: Assisted the logistics for Indian Space Congress, managing 150+ delegates and supporting research and session coordination

ACHIEVEMENTS

- 1st place at 10+ national debate tournaments across 15 states; chaired finals at 20+ tournaments including international as an invited adjudicator
- Honourable Mention, IMC (International Mathematics Competition) Olympiad for University Students 2022
- 1st Place across three national research conferences of IIT Delhi, IIM Indore, and SRCC.
- Discovered an asteroid through NASA's IASC programme using Pan-STARRS telescope data